

MDO Architecture Comparison

Architecture	Optimizer Level	MDA Required	Parallel Disciplines	Variable Count	Notes
MDF	Single	Yes (inner loop)	No	Low (n)	Simple; MDA cost dominates
Sequential	Hierarchical	Per discipline	No	Low-Med	Good for loosely coupled
IDF	Single	No	Yes	High ($n + \sum n_i$)	Coupling as constraint
CO	Bi-level	No	Yes	High	Each discipline independent
SAND	Single (large)	No (residuals)	Yes	Very high	Needs analytic gradients